

Research Advancement Plan

Fred

1 Background

High harmonic generation (HHG) is one of the most important processes driving the study of the interaction of radiation with matter and will lead to important advances in fundamental science and engineering. Through HHG, it is possible to produce coherent radiations that outperform the external in-field in regards to frequency and pulse duration.

The current theory is based on an approximation that ignores the very fact that acceleration of charged particles is in itself, a source of radiation. Such radiations as well as produced HHG effect on the whole system is neglected during the dynamics, among common other simplifications like the dipole approximation.

2 Objectives and methods

Guided by the work of Yangaliev *et al.* [1], we want to develop a reliable method of resolution of the inhomogeneous Schrödinger equation, taking into account the radiation emitted by the particle, at various levels. In other words, we want to build a robust model capable of comparing results ranging from the classical ansatz to the full quantum treatment of the radiation, but also from the strong field approximation to the numerical resolution of the Time Dependent Schrödinger Equation (TDSE), in the simplified context of 1D test cases before moving on to more realistic systems.

In order to ensure this transition, we will test our results against, some chosen references. Precandidates are suggested down below. Doing so will allow us to connect our work to the whole literature, still dominated by the work of Lewenstein.

3 Timetable

We shall meet the following deadlines

3.1 Validation of the main result

- **2026/4/31**

Producing the first full quantum compliant results described by Yangaliev *et al.* [1], for some emission frequencies only. (not the full spectrum)

- **2026/5/31**

Producing of the full quantum compliant HHG spectra, with converged results in the case of the field strengths $F_0 = 0.03$, and $F_0 = 0.1$ a.u., for both frequencies $\omega = 0.114$ and 0.57 a.u, described by Yangaliev *et al.* [1].

Notes

This will fully validate our model and allow us to reproduce the results displayed in the section IV of the reference to a satisfactory agreement. Converged results will easily be produced for the lesser intensities, but the higher ones will realistically take time and will be researched during the subsequent months.

3.2 The adiabatic approximation

- **2026/6/31**

In this part, we will study and compare our work to Tolstikhin *et al.* [2]. The interest of this reference for us is twofold

- it will allow us to understand the Adiabatic Approximation (AA) and give us a strong foothold for both the development of a resolution technique of the TDSE that is not a brute force numerical resolution
- it will but also will prepare us to our next step which is the extension to realistic 3D systems.

Notes

(N1) The AA will allow us to produce alternative results that may also directly compare to Strong Field Approximation (SFA) results rampant in the literature for strong external fields.

(N2) The AA will also allow us to design internal experiments for assessing our new fully quantum model in both strong and weak fields.

3.3 Extension to realistic systems

- **2026/9/31**

At this stage, we will have a functional fully quantum algorithm of resolution of the TDSE, but also an alternative semi-classical method in the AA. We will aim at extending our work to realistic systems. Following Tolstikhin *et al.* [3], who already successfully accomplished this by extending [2], to 3D systems with finite range potentials, we will have comparing points for our results, but most importantly a pathway for our approach.

Notes

(N1) After this point, we will have a two fully functional resolution mechanisms for the TDSE, allowing not only to compare ourselves to the general literature but also to make internal comparison experiments.

- **2026/10/31**

Last verifications, checks and convergence parameters tuning. If necessary, comparison of our results to other works in the literature for final validation or, following Linblom *et al* [4], extension of our work to rotating external fields.

- **2026/11~**

Writing a paper and the thesis manuscript.

Primary references

- [1] D. N. Yangaliyev, V. P. Krainov and O. I. Tolstikhin, Phys. Rev A 101, 013410 (2020)
- [2] O. I. Tolstikhin, T. Morishita, and S. Watanabe, Phys. Rev. A 81, 033415 (2010)
- [3] O. I. Tolstikhin, and T. Morishita, Phys. Rev. A 86, 043417 (2012)
- [4] T. K. Lindblom, O. I. Tolstikhin and T. Morishita, Phys. Rev. A 104, 023110 (2021)
- [5] M. Ohmi, O. I. Tolstikhin and T. Morishita, Phys. Rev. A 92, 044302 (2015)